

BUILDING ENERGY + HVAC DIGITAL TWIN

Vendor-neutral, calibratable open-source R&D reference model | MIT | vicena.ai

Transparent physics

Single-zone 1R1C heat balance

Calibration

Bounded least squares with uncertainty

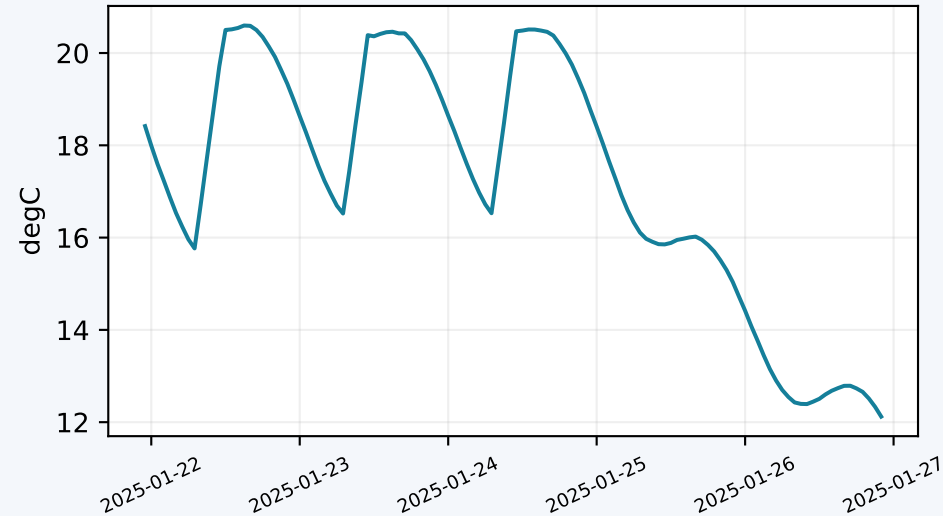
Validation

Temporal held-out metrics and residuals

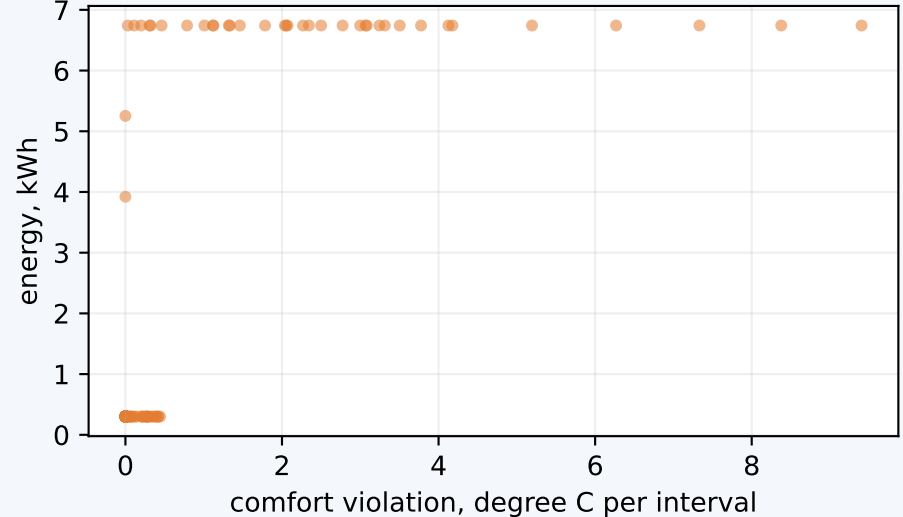
Optimization

Cost and comfort setpoint search

Held-out predicted indoor temperature, synthetic example



Energy and comfort outcomes, synthetic example



VERIFIED SYNTHETIC REFERENCE METRICS

503 calibration rows | 217 held-out rows | 0.442 degC temperature RMSE | 50.8% energy CVRMSE

Workflow: validate data > split > calibrate > held-out validation > sensitivity and constrained optimization